

MID-SEM-Question-07

● Graded

Student
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Total Points
10 / 12 pts

Question 1

Q7(a)2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect/not attempted

Question 2

Q7(b)2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect/not attempted

Question 3

Q7(c)2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect/not attempted

1A is not closed unless x is in A!!!

Question 4

Q7(d)2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect/not attempted

Question 5

Q7(e)2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect/not attempted

+ 2 pts Correct

✓ + 0 pts Incorrect/not attempted

Name:

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Roll No:

241211

• Write your name cleanly in CAPITAL letters and roll number inside the boxes.

Solution of Q7 (a).

No, take a ~~continuous~~ onto ^{one} map $f: [0,1] \rightarrow (0,1)$

now consider the restriction on domain of $[0,1]$ and remove $c = f(0)$ from $(0,1)$ then the map ~~is~~

$f: [0,1] \rightarrow (0,c) \cup (c,1)$ is still onto but this cannot be continuous

as it maps connected set to disconnected set (if f is onto & continuous it maps connected set to connected set)

as f cannot be continuous on $(0,1)$ hence $[0,1]$ it cannot be a homeomorphism.

Solution of Q7 (b).

~~No~~ suppose f is homeomorphism $f: (0,1) \rightarrow [0,1]$

we know if E is open in $(0,1)$ then $f(E)$ is open in $[0,1]$

by similar logic take an onto map $f: [0,1] \rightarrow (0,1)$ and consider its restriction on $(0,1)$, remove $c = f(0)$ & $d = f(1)$ from $(0,1)$

w.l.o.g $C \leq d$.

$f: [0,1] \rightarrow (0,c) \cup (c,d) \cup (d,1)$ is onto again this cannot be continuous as it maps connected set to disconnected set

$\Rightarrow f$ cannot be homeomorphism (cannot have onto map which is continuous)

Solution of Q7 (c).

false.

$\{x_n\} \in M$ let $|x_n| > n$

$\nexists \{x_{n_k}\} \in M$ s.t. $x_{n_k} \rightarrow x \in M$.

~~as~~ take $A = \{x_{n_k} : k \geq 1\}$

~~f is c.~~ $\Rightarrow A$ is compact (closed subset of compact space is compact)

f is continuous

$\Rightarrow f(A)$ is compact

Solution of Q7 (d).

true,

take $x_1, x_2 \in \mathbb{Q}; x_1 < x_2$ by density of irrationals $\exists y \in \mathbb{R} \setminus \mathbb{Q}$ s.t. $x_1 < y < x_2$

$\Rightarrow \mathbb{Q} \subseteq (-\infty, y) \cup (y, \infty)$ also $(-\infty, y) \cap \mathbb{Q} \neq \emptyset$ (contains x_1)

~~$(-\infty, y) \cap \mathbb{Q} \neq \emptyset$~~ $(y, \infty) \cap \mathbb{Q} \neq \emptyset$ (contains x_2)

$\Rightarrow (-\infty, y)$ & (y, ∞) are open disjoint sets and hence form a disconnection on $\mathbb{Q}$

$\Rightarrow \mathbb{Q}$ is disconnected.

Solution of Q7 (e).

false.

take ~~the~~ set d , to be discrete metric

and $A = [0,1] \subseteq \mathbb{R}$

A is closed (any subset of discrete space is closed)

A is bounded.

~~$B_\epsilon(0)$~~ $A \subseteq B_2(0)$

but A is not totally bounded.

as A , take $\epsilon = \frac{1}{2}$.

~~$B_\epsilon(x) = \{x\}$~~ to cover entire A we need infinite balls

$\Rightarrow A$ is not totally bounded hence not compact.

Solution of Q7 (f).

yes.

~~A is total~~
~~A is closed~~ (as x_{n_k} is Cauchy)
1

$\Rightarrow f(x_{n_k})$ has a convergent subsequence
but $|f(x_n)| > \epsilon \forall n$ hence contradiction